Detecting and Interpreting Text Generated by Large Language Models: A Literature Review

Deogracias Mufula
StudentNumber:MFLDEO001
Department of Computer Science, University of Cape Town

Abstract

Large language models have made it harder to distinguish between AI-generated text from human writing, posing serious problems with content authenticity, academic integrity, and disinformation. The area of LLM-generated text detection is examined in this study, which covers important detection techniques, evaluation benchmarks, and the real-world constraints that make existing systems unreliable. This project's primary goal is to close the gap between interpretability and detection performance.

Keywords

LLM-generated text detection, machine-generated text, zero-shot detection, log-likelihood, natural language processing, interpretability

1 Introduction

The recent development of Large Language Models (LLMs) such as GPT-3, GPT-4, and ChatGPT has changed the scope of text production. These models are capable of generating contextually coherent, human-like text at scale [12]. Academic integrity, journalism, political influence operations, targeted propaganda, and internet spam are just a few of the areas where this capability has serious repercussions. There is an urgent need for systems that can consistently differentiate compelling machine-generated writing from human-written content due to the simplicity with which it can now be produced.

The core challenge this creates is one of verification: given a piece of text, determining whether it was produced by a human or a language model. Detectors must deal with a wide variety of generative models, writing domains, languages, and adversarial conditions, none of which are uniformly handled by existing approaches [12]. No single method has proven reliably effective across these axes, and the gap between controlled benchmark performance and real-world reliability remains one of the defining open problems in the field [13].

This literature review surveys the landscape of LLM-generated text detection with the aim of mapping that gap. It covers five categories of work: early approaches that established foundational baselines, zero-shot statistical methods that utilise probability distributions without labeled data, watermarking methods that embed detectable signals at generation time, supervised neural methods that learn to classify from labeled examples, and the benchmarks and evaluation frameworks used to assess them all [12, 13]. Each category is examined in terms of its theoretical foundations, empirical performance, and practical constraints.

A gap that runs across this landscape is the tension between detection performance and interpretability: the highest-performing methods offer no insight into why a text was flagged, while interpretable methods sacrifice robustness. This review maps that gap and positions log-likelihood thresholding as a method that balances between the 2.

2 Early Approaches to Machine-Generated Text Detection

2.1 Neural Fake News and Generator-Based Detection

The problem of detecting machine-generated text predates the current era of LLMs, with early work focused on model-generated fake news. Zellers et al. [15] introduced GROVER, a transformer-based model capable of generating coherent, convincing news articles with only a headline, author, and publication outlet given. The motivation was that by building a powerful generator, the authors aimed to understand and counter the threat of neural fake news at scale.

The central finding of Zellers et al. [15] was as follows: the most effective detector of GROVER-generated text was GROVER itself. GROVER was trained as a conditional language model on a large corpus of news articles, learning to generate text conditioned on metadata such as the headline, author, and publication domain. In order to repurpose it as a detector, the same model was used to score candidate articles by computing the likelihood of the text under its learned distribution, with generated text scoring higher than human-written text. When used in this way it achieved 92% accuracy in distinguishing its own output from human-written news, outperforming other architectures including GPT-2 and BERT, whose accuracy declined as GROVER's model size increased. Human participants, achieved only slightly above 50% accuracy when distinguishing between human-written and machine-generated text, indicating performance close to random guessing.

This result established two foundational insights. First, that the generator and the detector are deeply linked. A model that understands how to produce convincing text is also best positioned to detect it. Second, that human judgment alone is insufficient as a detection mechanism, motivating the need for automated approaches.

2.2 Human Ability to Detect Machine-Generated Text

Ippolito et al. [4] investigated this second insight directly by conducting a systematic study of human ability to detect machine-generated text across different decoding strategies. Their central finding was counterintuitive: text that humans rated as higher quality was actually harder for them to identify as machine-generated, yet easier for automated detectors to flag. This suggests that as LLMs improve, the gap between human and automated detection

capability will widen. The study also found that exposing human evaluators to examples before assessment improved their accuracy, mainly on longer texts, but that this improvement was inconsistent. Ippolito et al. [4] concluded that human judgment cannot be relied upon as a primary detection mechanism, reinforcing the case for automated approaches while also raising a deeper concern: if the best human-generated text and the best machine-generated text are indistinguishable to humans, the burden falls entirely on statistical and model-based methods.

A key limitation of both GROVER and the Ippolito et al. [4] study is that they were conducted in a narrower technological landscape. GROVER's detection ability was closely tied to the distribution of its own generated outputs, and the texts evaluated by Ippolito et al. were produced by earlier, weaker models. Their findings establish important baselines but do not generalise directly to the outputs of modern LLMs such as GPT-4, a weakness later systematically demonstrated by benchmarks such as DetectRL [13].

3 Zero-Shot Statistical Detection Methods

Zero-shot statistical methods represent an alternative to supervised detection, operating without labeled training data by exploiting measurable properties of how language models generate text. Rather than learning to classify from examples, these methods derive detection signals directly from the probability distributions of LLMs, making them theoretically flexible and broadly applicable across models and domains, a flexibility that earlier model-specific approaches such as GROVER could not achieve.

Solaiman et al. [10] established the foundational baselines for this category, introducing log-likelihood and rank as zero-shot detection signals. Log-likelihood measures the average per-token log probability that a language model assigns to a given text. Formally, given a text $x = (x_1, x_2, \ldots, x_n)$, the average per-token log-likelihood under a language model p_θ is defined as:

$$\mathcal{L}(x) = \frac{1}{n} \sum_{i=1}^{n} \log p_\theta(x_i \mid x_1, \ldots, x_{i-1}) \tag{1}$$

LLM-generated text tends toward higher values of $\mathcal{L}(x)$ because the model produces tokens that are statistically likely given the preceding context [8, 10]. Rank measures where each token falls in the model's output probability distribution, with LLM-generated text tending toward lower average ranks. These two metrics, simple as they are, remain competitive baselines against which more sophisticated methods are still benchmarked today, underlining how much detection signal is available from raw probability information alone.

Gehrmann et al. [2] extended this idea into a visual forensic tool called GLTR(Giant Language Model Test Room), which highlights tokens in a text according to how likely they are under a language model's distribution. Rather than producing a single classification score, GLTR makes the statistical signal interpretable to human reviewers, colouring tokens by sampling frequency so that evaluators can see where a text is suspiciously probable.

The tool demonstrated that even non-expert humans could improve their detection accuracy when given access to this visual signal, achieving significantly better performance than unaided judgment.

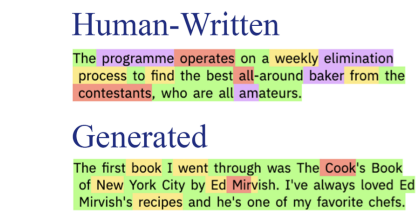


Figure 1: GLTR colour-coded token overlay on human-written and GPT-2 generated text. Adapted from Gehrmann et al. [2], Figure 1.



Figure 2: Left: a GPT-2 generated sample annotated with GLTR, showing the generated text mostly highlighted in green and yellow, strongly hinting at automatic generation. Right: a real NYT article and a scientific abstract, where the fraction of red and purple words is much higher. Adapted from Gehrmann et al. [2], Figure 3.

However, GLTR was designed as an assistive tool rather than a standalone classifier, and its effectiveness depends on the reviewer's ability to interpret the visualisation correctly, a limitation that reduces its scalability. Mitchell et al. [8] introduced DetectGPT, one of the most influential zero-shot detection methods in the literature. The core idea is that LLM-generated text tends to occupy regions of high log-probability in the model's distribution, that is, the model assigns high likelihood to its own outputs. Human-written text, by contrast, does not appear in these high-probability regions with the same consistency.

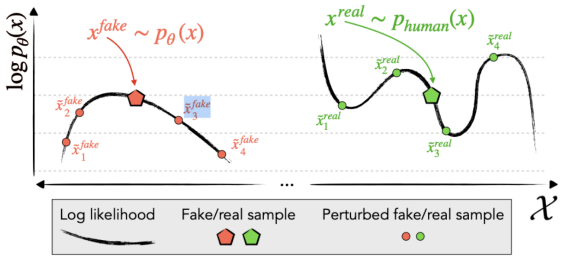


Figure 3: Machine-generated text tends to occupy negative curvature regions of the log probability function, where nearby samples have lower model log probability on average. Human-written text does not exhibit this pattern consistently. Adapted from Mitchell et al. [8], Figure 2.

DetectGPT utilises this observation by comparing the log-probability of a candidate text against the log-probabilities of slightly perturbed versions of that same text, generated using a mask-filling model such as T5. If the original text has consistently higher log-probability than its perturbations, it is likely machine-generated. The perturbation-based score used by DetectGPT is defined as:

$$d(x, p_\theta, q) = \log p_\theta(x) - \mathbb{E}_{\tilde{x} \sim q(\cdot|x)} \log p_\theta(\tilde{x}) \tag{2}$$

where $q(\cdot|x)$ is the perturbation model (T5) and $\tilde{x}$ are perturbed versions of x [8]. On its benchmark evaluations, DetectGPT achieved strong AUROC scores, outperforming prior zero-shot baselines including the log-likelihood and rank methods of Solaiman et al. [10].

However, DetectGPT carries practical limitations. The perturbation step is computationally expensive, requiring multiple forward passes through both the scoring model and the perturbation model for each candidate text. Bao et al. [1] addressed this directly with Fast-DetectGPT, replacing the perturbation step with a conditional probability curvature approach that samples from the model's own distribution rather than generating external perturbations. Fast-DetectGPT achieved a 340-fold increase in detection speed while improving AUROC by approximately 75% over DetectGPT in controlled settings. Despite this improvement, Wu et al. [13] demonstrate through the DetectRL benchmark that both DetectGPT and Fast-DetectGPT degrade severely under real-world conditions, with DetectGPT achieving only 22.15% AUROC on academic writing and 12.21% on news writing, and Fast-DetectGPT scoring only 55.33% average across all tasks.

Xu et al. [14] approached zero-shot detection from a different angle, proposing a training-free method based on mining token probability sequences. Rather than relying on comparing among perturbations, the method identifies sequential patterns in the per-token probabilities assigned by a language model to a given text, exploiting the observation that LLM-generated text exhibits more consistent and structured probability sequences than human-written text. The method was evaluated across multiple LLMs and domains.

Altogether, these methods illustrate both the capabilities and limitations of zero-shot statistical detection. The progression from raw log-likelihood through GLTR and DetectGPT to Fast-DetectGPT reflects steady refinement of the core probability-based signal, while the DetectRL results make clear that real-world robustness remains an open problem for all of them. Both methods also depend on access to model probabilities, which limits their applicability when analysing outputs from closed-source, black-box LLMs such as GPT-4, where logits are not directly accessible.

4 Watermarking Methods

Watermarking represents a fundamentally different approach to LLM-generated text detection, one that operates proactively rather than reactively. Rather than analysing a piece of text after the fact, watermarking methods embed a detectable signal into the text during the generation process itself, making it possible to verify the origin of a text without requiring access to external classifiers or probability comparisons.

Kirchenbauer et al. [5] introduced the first practical watermarking framework for LLMs, and it remains the most widely cited approach in the literature. The method works by partitioning the model's vocabulary into two sets before each token is generated, a "green" list and a "red" list, determined by a hash of the preceding token. During generation, the model is slightly guided to favour green-list tokens. The resulting text carries a statistical watermark that can be detected by any party with knowledge of the hash function, without requiring access to the model's weights or logits. On benchmark evaluations, the method achieved reliable detection with minimal impact on text quality, and the authors demonstrated that the watermark could be detected using a simple statistical test based on the proportion of green-list tokens in the text. Detection is performed using a one-sided z-test:

$$z = \frac{|s|_G - \gamma T}{\sqrt{T\gamma(1-\gamma)}} \tag{3}$$

where $|s|_G$ is the count of green-list tokens, T is the total token count, and γ is the green list fraction [5].

Critically however, the robustness of this approach has been contested. Kirchenbauer et al. [5] themselves acknowledge that the watermark can be disrupted simply by paraphrasing, and subsequent work has shown that synonym substitution and back-translation can reduce detection reliability significantly. A further practical limitation is that watermarking requires cooperation from the LLM provider: the watermark must be embedded at generation time, meaning it cannot be applied retroactively to text that has already been produced. This makes watermarking ineffective for detecting text generated by open-source models where the generation process is not controlled, or for text that predates the watermarking system. Wu et al. [12] also notes that watermarking methods are entirely inapplicable to closed-source black-box models unless the provider implements the scheme themselves, which limits their real-world coverage considerably.

Despite these limitations, watermarking occupies an important theoretical position in the detection landscape. Unlike statistical and supervised methods, it does not rely on any observable difference between human and machine-generated text.

5 Supervised Neural Detection Methods

Supervised neural detection methods represent the strongest performing category in the literature, consistently outperforming zero-shot approaches across benchmarks and real-world evaluations. Rather than exploiting probability distributions directly, these methods learn to classify text from labeled examples, typically by fine-tuning pre-trained language models on paired human-written and LLM-generated text.

The dominant approach is fine-tuning RoBERTa (Robustly Optimized Biderectional Encoder Representations for Transformers Approach) [16], a pre-trained transformer model. Wu et al. [12] report that fine-tuned RoBERTa classifiers achieve approximately 95% accuracy within their training domain, outperforming both zero-shot and watermarking methods under controlled conditions. Wu et al. [13] corroborate this, with RoBERTa-Base topping the DetectRL leaderboard at 93.02% average performance, significantly ahead of the best zero-shot method, Binoculars, at 79.61%.

The literature has also explored extensions to improve robustness, including contrastive learning to address data scarcity and

adversarial training frameworks such as RADAR and OUTFOX, which harden detectors against paraphrase and substitution attacks [12].

Critically however, supervised methods tend to overfit to their training distribution. Classifiers trained on one LLM's output frequently fail on another, and performance drops meaningfully across domains [13]. A further concern is that as LLM-generated text floods the web, supervised detectors risk being trained on the very content they are meant to detect, undermining their reliability [12].

Finally, using LLMs themselves as detectors has proven unreliable. GPT-4 misclassifies roughly 95% of human-written text as LLM-generated, while ChatGPT exhibits the opposite bias [12], making LLM-based detection unsuitable as a standalone approach.

6 Benchmarks and Evaluation

The reliability of any detector is only as meaningful as the evaluation framework used to assess it. A recurring problem in the literature is that detectors are frequently evaluated on idealised test sets that do not reflect real-world conditions/complexities, producing performance figures that overstate practical utility [12]. The benchmarks reviewed here represent progressively more sophisticated attempts to address this gap, moving from single-domain paired datasets through multilingual benchmarks to comprehensive real-world simulation frameworks.

Guo et al. [3] introduced HC3, one of the first large-scale datasets explicitly designed to compare human and ChatGPT responses to identical questions. The dataset contains approximately 58,000 human responses and 26,000 ChatGPT responses in English, drawn from sources including Wikipedia, medical dialogue datasets, and question-answering forums, with a parallel Chinese dataset covering seven additional domains. HC3 established a foundational resource for training and evaluating detectors in the post-ChatGPT era, and its paired question-answer structure made it particularly useful for studying how ChatGPT-generated text differs from human responses in tone, length, and vocabulary. Critically however, HC3 has notable limitations: its prompts lack diversity, it covers only ChatGPT as a generative model, and its construction predates the adversarial and multi-model challenges that subsequent benchmarks were designed to address.

Wang et al. [11] introduced M4, a comprehensive multi-generator, multi-domain, and multi-lingual benchmark that significantly broadened the evaluation landscape. M4 spans more than ten languages and incorporates text generated by models including ChatGPT, LLaMA, BLOOMz, FlanT5, and Dolly, drawn from sources such as Wikipedia pages, news outlets, and academic portals. Its multi-generator design was a direct response to the limitation of earlier benchmarks like HC3, which focused on a single model. M4 demonstrated that detection performance varies substantially across generators and languages, with detectors trained on one model's output frequently failing on another's. A limitation noted by Wu et al. [12] is that M4, despite how broad it is, still reflects relatively idealised generation conditions and does not systematically evaluate adversarial scenarios.

Macko et al. [6] introduced MULTITuDE, a large-scale multilingual benchmark covering eleven languages including Arabic, Chinese, German, Russian, and Spanish, with approximately 74,000 machine-generated and 7,000 human-written texts produced by eight generative models. Its key finding was that detection performance varies substantially across languages, with detectors performing well on English but degrading on lower-resource languages. This exposed the cross-lingual fragility of existing detectors and established multilingual evaluation as a necessary dimension of any serious assessment.

Macko et al. [7] extended this with MultiSocial, targeting machine-generated text specifically in social media contexts across multiple languages. Social media text is shorter, more informal, and more fragmented than the academic and news writing that dominates prior benchmarks, and MultiSocial surfaces a category of real-world text that existing detectors handle poorly.

Wu et al. [13] introduced DetectRL, the most comprehensive real-world benchmark to date. Rather than varying only domain and language, DetectRL simulates conditions under which detectors most commonly fail in practice, including adversarial prompt strategies, human revision of LLM-generated text, spelling errors, and mixed human-LLM content. It was constructed using four widely used black-box LLMs across academic, news, creative, and social media domains. The results were that adversarial perturbation attacks reduced zero-shot detector performance by an average of 39.28% AUROC, and DetectGPT, which had reported strong controlled results, achieved only 34.43% average AUROC, worse than several simple baselines.

Taken together, these benchmarks reveal a consistent pattern of progress and persistent limitation. HC3 established the paired comparison paradigm, M4 expanded it across models and languages, MULTITuDE exposed cross-lingual fragility, MultiSocial exposed informal domain weakness, and DetectRL exposed adversarial vulnerability. Their convergence across independent studies strengthens the case that the field needs not only better detectors but fundamentally more realistic evaluation frameworks.

7 Challenges and Limitations

The preceding sections reveal a consistent pattern: detection methods that perform well in controlled settings frequently fall short in practice. This section consolidates the core challenges that cut across all approaches, drawing primarily on Wu et al. [12] and Wu et al. [13].

The most pervasive challenge is generalisation. Detectors trained on text from one domain or one LLM reliably degrade when applied to a different distribution. Wu et al. [13] quantify this through DetectRL, showing performance drops of up to 20% when moving across domains, suggesting that differences in model training produce statistical patterns that existing detectors are not built to handle.

Adversarial attacks represent a second critical vulnerability. Wu et al. [12] identify three main attack types: paraphrase attacks, perturbation attacks, and prompt attacks. Perturbation attacks are the most damaging, reducing zero-shot detector performance by an average of 39.28% AUROC in DetectRL. Back-translation, a common paraphrasing strategy, is similarly effective. Supervised detectors are more resilient but still vulnerable, particularly when attack types at test time differ from those seen during training.

A deeper theoretical challenge is raised by Sadasivan et al. [9], who argues that reliable detection may be fundamentally impossible under certain conditions. Their analysis demonstrates that as LLM-generated text becomes statistically closer to human-written text, the total variation distance between the two distributions approaches zero, meaning that no detector, regardless of its design, can achieve low error rates on both classes simultaneously. Under paraphrase attacks specifically, they show that the performance of any detector is bounded by the quality of the paraphraser, and that sufficiently powerful paraphrasers can reduce detection to near-random accuracy. This is a significant theoretical result because it suggests that the robustness problems documented by Wu et al. [13] are not merely engineering limitations but may reflect a fundamental constraint on what detection can achieve.

A further challenge is mixed and human-edited text. In practice, much of the text detectors will encounter is not purely LLM-generated, as it may combine human and machine-written content, or human text revised by an LLM. Wu et al. [12] identify this as one of the most pressing unsolved problems in the field, noting that current detectors are designed for the binary case and lack the granularity for sentence or word-level attribution.

Finally, the field suffers from a lack of standardised evaluation. Reported performance figures are difficult to compare across studies because researchers use different datasets, generation parameters, and pre-processing strategies. Wu et al. [12] note that this reproducibility problem means strong reported numbers often say more about evaluation design than detector quality, a concern that DetectRL was explicitly built to address.

For this project, these challenges serve as a frame. These limitations apply equally to the classifier implemented in this project, and confronting them directly is central to producing a meaningful analysis of its performance.

8 Discussion

This project addresses that need directly by implementing a binary classifier for machine-generated text detection based on log-likelihood thresholding, which is a zero-shot statistical method that requires no labeled training data and exploits a fundamental property of how language models generate text. The method is grounded in the observation that language models produce text by sampling from high-probability regions of their distribution, leaving behind a detectable statistical signature in the form of elevated average per-token log probability. This signal differs measurably from human-generated text, where word choice is driven by meaning and context rather than statistical likelihood [8].

Understanding this distinction is essential before implementation, as it clarifies both what the classifier is measuring and where its limitations lie. This matters because the ability to reliably distinguish human-written from AI-generated text has direct real-world impact for academic integrity, disinformation, and public trust in online content [13].

With that framing in place, the literature surveyed in this review reveals a field that has advanced rapidly in technical sophistication but has not yet resolved its most fundamental tension: the gap between controlled benchmark performance and real-world reliability. This tension runs through every category of detection method reviewed, and it has direct implications for how the classifier implemented in this project should be understood and evaluated.

Zero-shot statistical methods, including the log-likelihood approach adopted here, offer a theoretically principled basis for detection that does not depend on labeled training data or a fixed model distribution. This is a meaningful practical advantage over supervised methods, which overfit to their training distribution and degrade when applied to unseen LLMs or domains. However, the DetectRL results make clear that zero-shot methods are not robust by default. Their performance is sensitive to domain style, text length, and adversarial perturbation, and the gap between their best-case and worst-case performance is substantial. Log-likelihood thresholding shares these vulnerabilities. Its detection signal is strongest when the scoring model closely matches the generative model, and weakest when applied across domains or to text that has been post-processed.

Supervised methods, while more robust in practice, introduce a different set of problems. Their dependence on labeled data, their tendency to overfit, and the growing risk of training on contaminated web data all limit their long-term reliability. More critically for this project, they offer no interpretable signal. A fine-tuned RoBERTa classifier produces a probability score with no indication of which features of the text drove the decision. Log-likelihood thresholding, by contrast, operates at the token level, making the detection signal inherently more transparent and traceable.

The theoretical work of Sadasivan et al. [9] introduces a harder boundary on what detection can achieve. Their result that the total variation distance between human and LLM-generated text distributions approaches zero as models improve implies that no detector, including log-likelihood thresholding, can maintain reliable performance indefinitely as LLMs advance. Under strong paraphrase attacks, they demonstrate that detection degrades toward random accuracy regardless of the method used. This is an important constraint on how their results should be interpreted. The value of a log-likelihood classifier is not that it solves detection permanently, but that it provides a transparent and theoretically grounded signal that can be critically analysed within these known limits.

The benchmark progression from HC3 through M4, MULTITuDE, and MultiSocial to DetectRL reflects a maturing understanding of what realistic evaluation requires. Each benchmark has expanded the axes along which detectors are tested, and each has revealed new failure modes. For this project, DetectRL is the most relevant reference point as its real-world simulation of prompt variation, human revision, and adversarial noise defines the conditions under which the log-likelihood classifier is most likely to be challenged. Understanding where it falls on the DetectRL performance spectrum is therefore more informative than any controlled benchmark result.

Comparing the four categories directly, supervised methods lead on raw performance but fail on interpretability and generalisation. Zero-shot methods, including log-likelihood thresholding, sacrifice peak accuracy for flexibility and transparency — they require no labeled data, degrade gracefully across domains, and produce a signal that can be inspected at the token level. Watermarking is the only proactive approach, but its dependence on provider cooperation makes it inapplicable to the majority of real-world detection scenarios. Across all three, the theoretical result of Sadasivan et

al. [9] applies equally: no method is immune to the fundamental distributional convergence problem as models improve. The choice of log-likelihood thresholding for this project is therefore not a claim of superiority but a deliberate trade-off — prioritising interpretability and zero-shot flexibility within known theoretical limits.

Across these studies, the literature positions log-likelihood thresholding as a method with clear theoretical grounding, practical accessibility, and interpretive transparency, but with well-documented limitations in robustness and generalisation. These limitations are not unique to this method as they reflect the state of the field as a whole. The value of this project therefore lies not in delivering a state-of-the-art detector, but in engaging critically and precisely with what this method measures, where it holds up, and what its failure modes reveal about the broader and still unsolved challenge of distinguishing human from machine-generated text.

9 Conclusions

This review has surveyed the landscape of LLM-generated text detection across five categories: early approaches, zero-shot statistical methods, supervised neural methods, benchmarks and evaluation, and the challenges that cut across all of them. Several conclusions can be drawn.

First, no single detection approach is universally reliable. Zero-shot methods are flexible but break down under adversarial conditions and across different domains. Supervised methods perform strongly in controlled settings but struggle when applied outside their training data. The gap between reported performance and real-world reliability, documented most clearly by Wu et al. [13], remains one of the defining problems of the field.

Second, how detectors are evaluated matters as much as the detectors themselves. The progression from early benchmarks through MULTITuDE and MultiSocial to DetectRL reflects a growing recognition that overly controlled test conditions produce results that do not hold up in practice. Meaningful evaluation must account for domain variation, adversarial attacks, and the complexity of real-world text.

Third, the field has prioritised accuracy over transparency. The best performing detectors offer no insight into why a piece of text was flagged as machine-generated, which is a real problem in high-stakes settings such as academic integrity, where a decision with no explanation is difficult to trust or challenge.

These conclusions directly motivate this project. By implementing a log-likelihood based classifier, this project works with a method that is theoretically grounded, does not require labeled training data, and produces a detection signal that is more transparent than neural classifiers. The challenges identified in this review will serve as the lens through which the classifier's results are analysed and not to claim it solves what the field has not, but to understand clearly where it works, where it does not, and what that tells us about the broader problem of detecting machine-generated text.

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